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Fuzzy trust-based secure localization with CNN intrusion detection and RANSAC-GDA for wireless sensor networks

Aparna Singh[1], Surabhi Solanki[1†], Surbhi Sharmai[1†], Pulakesh Roy[2†], Susmita Das[1†] and Rahul Sharma[3*]

†Surabhi Solanki, Surbhi Sharmai, Pulakesh Roy, Susmita Das and Rahul Sharma contributed equally to this work.

*Correspondence:
Rahul Sharma
sharma.rahul@jaipur.manipal.edu; Sharma.rahul5656@gmail.com
[1]School of Computer Science Engineering and Technology, Bennett University, Greater Noida, India
[2]Department of Computer Science Engineering, Brainware University, Kolkata, India
[3]Department of Artificial Intelligence and Machine Learning, Manipal University Jaipur, Jaipur, India

Abstract

Wireless Sensor Networks (WSNs) have made possible many applications like environment tracking, healthcare, smart cities, etc. However, reliable node localization remains challenging due to energy constraints, noisy range estimates, and malicious adversaries. This paper proposes a hybrid secure and energy-efficient localization framework that integrates Distributed Energy-Efficient Clustering (DEEC), Random Sample Consensus (RANSAC), and a Gaussian Distribution Approach (GDA) to enable robust position estimation. In addition, a lightweight Intrusion Detection System (IDS) based on a SMOTE-assisted Convolutional Neural Network (CNN) is deployed at the pre-localization stage to identify and exclude malicious nodes, including Sybil behavior and falsified localization information. DEEC-RANSAC-GDA facilitates energy-efficient selection for cluster head formation, mitigates localization outliers via RANSAC, and optimizes localization using Gaussian probabilistic models. Implementation of the proposed IDS on the CIC-IDS-2017 benchmark dataset achieved 99.83% accuracy, 99.00% precision, 99.84% recall, and 99.42% F1-score, along with minimizing localization errors falling within the range of 0.46 m-0.53 m compared to DV-Hop and Weighted Hyperbolic Localization schemes.

Keywords Localization, Intrusion detection, SMOTE, DEEC, RANSAC, Outlier detection

1 Introduction

Wireless Sensor Networks (WSNs) have attracted much attention in recent times due to their diverse range of applications in environmental monitoring [1], healthcare [2], disaster management [3], and smart cities [4]. Accurate localization is crucial in WSNs for efficient data routing and energy management. However, challenges such as environmental noise and malicious nodes compromise localization accuracy, making it a persistent research problem.

Localization methods typically rely on distance estimation techniques. However, these techniques are prone to inaccuracies due to noise, obstacles, and malicious nodes that tamper with data. To overcome these challenges, efficient localization algorithms have been designed to integrate energy-efficient clustering mechanisms and outlier detection

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techniques. This article proposes a hybrid localization model that uses the Distributed Energy-Efficient Clustering (DEEC) protocol with the Gaussian Distance Algorithm (GDA) to improve accuracy, and Random Sample Consensus (RANSAC) to enhance localization precision. One of the main threats to localization accuracy in WSNs is malicious nodes that might offer fake distance or coordinate information, resulting in wrong localization results. Our proposed methodology mitigates this threat by including an Intrusion Detection System (IDS) using Convolutional Neural Networks (CNN) and utilizing a Synthetic Minority Oversampling Technique (SMOTE). An IDS preprocesses before the localization process to identify and exclude malicious nodes. This prevents malicious data usage, making the algorithm localization more reliable.

The DEEC algorithm prolongs the network's lifespan by dynamically adjusting the selection probability of the cluster head (CH) based on residual energy, ensuring proper communication among nodes [5]. In the proposed method, DEEC is further enhanced by integrating the GDA and RANSAC algorithms to improve localization performance [6]. GDA accounts for geometric factors that influence distance estimation, reducing the impact of unfavourable node placements. Meanwhile, RANSAC robustly identifies and excludes outliers, ensuring that erroneous measurements do not skew the localization process.

Figure 1 depicts a WSN with two categories of nodes: anchor and unknown sensor nodes. Anchor nodes have their positions known, either because of the GPS-based localization or even by manual geo-positioning before deployment. The anchor nodes provide reference points in localization. Sensor nodes that are unaware of their location are referred to as unknown sensor nodes.

Some recent studies have explored localization optimization with secure localization under adversarial settings. However, most of these schemes remain hypersensitive to noise, anchor sparsity, and outliers. Also, the intrusion detection framework explored recently does achieve higher accuracy and precision, but its integration with localization still remains unexplored. Accordingly, the major research gaps that motivated this work include limited work on energy-aware clustering with robust outlier removal and

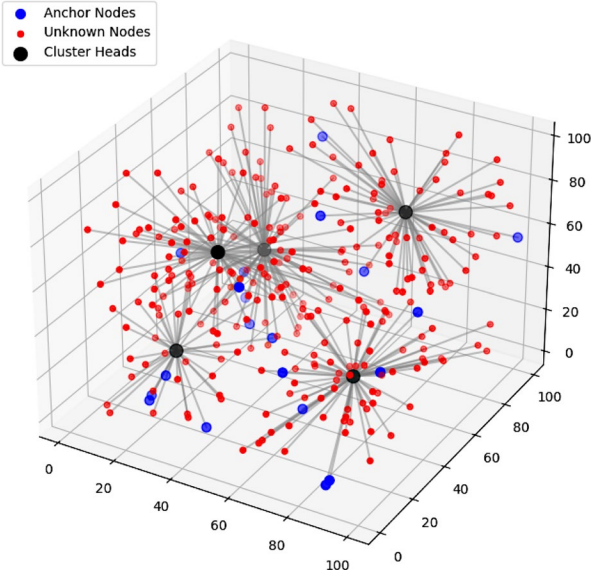


Fig. 1 3D model of a sensor network incorporating anchor and unknown nodes

probabilistic localization refinement, all processed within a single localization pipeline. Furthermore, insufficient analysis of the effect of malicious nodes on localization accuracy lays the foundation of the proposed work. This work addresses these gaps by integrating DEEC-based clustering, RANSAC-based outlier suppression, Gaussian error modeling for localization refinement, and a lightweight CNN-based IDS at the pre-localization stage.

This article proposes a localization model integrated with a pre-IDS phase. Therefore, this architecture eliminates any malicious node's influence over the localization process, thereby improving the accuracy and reliability of node localization. The key contributions of this research include:

1. A hybrid localization framework combining DEEC-based clustering, RANSAC-based outlier suppression, and Gaussian error modeling is proposed to improve localization robustness and energy efficiency in WSNs.
2. IDS is integrated at a pre-localization step, based on an SMOTE-assisted CNN that enables filtering of malicious nodes before localization, thereby improving reliability under adversarial conditions.
3. A thorough result analysis provided in the end, using IDS classification metrics and localization error comparisons against DV-Hop and Weighted Hyperbolic methods under varying anchor configurations proves the efficiency of the proposed pipeline.

Unlike the existing trust-based WSN routing schemes which rely mostly on node-level fuzzy trust, without intrusion detection, this work fuses IDS confidence, residual energy and ranging consistency into a single fuzzy trust score combining it with RANSAC-GDA localization. This integration of filtering and localization into a single model is the core contribution of this work. The rest of the manuscript is outlined as follows. Section 2 includes a review of existing research on malware detection techniques in WSNs. It focuses on deep learning-based security measures and federated learning. Section 3 describes the proposed methodology, involving IDS, fuzzy system, outlier detection using RANSAC, and DEEC-GDA-based localization. Section 4 presents the performance of the proposed framework and provides a comparative analysis between the proposed model and existing techniques. The obtained results validate that the proposed model is a robust and efficient solution for intrusion detection in WSNs, with considerable improvements in security, privacy preservation, and localization accuracy. Lastly, Sect. 5 provides a summary of the proposed work while also covering the future scope of the proposed work.

2 Literature review

The integration of IoT devices with WSNs provides new opportunities for realising various applications from environmental monitoring to smart cities. But this has also raised the contest for attention related to security threats from the disruption that malicious nodes pose to network operations. To overcome these challenges, recent research has combined various evolutionary techniques to improve the security and reliability of IoT-based WSNs.

A detailed survey by Ammar et al. [7] has classified various secure localization schemes and approached counteractions that could endanger localization processes. It highlights how strong, rapid algorithms are an immediate requirement that must be

fortified with efficient and adequate resistance against malicious adversaries to uphold localization authenticity. The localization improvements have been further analysed in several studies.

Ahmad et al. [8] provided a detailed view of optimization algorithms for WSN node localization. Mani et al. [9] introduced a 3D localization algorithm designed for large-scale WSNs, which is highly accurate. This means that localization precision varies greatly; however, nothing has been specifically stated about mechanisms for treating malignant nodes. Similarly, Gumaida and Luo [10] suggested that a hybrid particle swarm optimization with variable neighbourhood search can improve localization accuracy. While it succeeds in reducing localization errors, it has been widely argued that it would be worthless in a hostile environment because of its security focus. Aroba et al. [11] used a hyper-heuristic DEEC-Gaussian gradient technique for node localization. This algorithm uses distance/proximity to attain higher accuracy. Neggaz et al. [12] determined that one could apply the Nutcracker optimizer for optimal node localization in WSNs. It is easier to use for the optimization problem, but it doesn't address the malignant activity detection or encryption of data being relayed into the WSN. Wu et al. [13] proposed a hybrid method to mobile node localization in wireless sensor networks in the paper, where their technique fuses the algorithms of Particle Swarm Optimization (PSO) and Monte Carlo Localization Boxing (MCB) to improve the three-dimensional, dynamic environment's node positioning accuracy. Since network conditions are always changing, this research focuses on the need for precise data representation for effective communication and functionality in such networks.Rayavarapu et al. [14] implemented an anchor-based localization technique using a hybrid electric-coyote optimizer, reinforcing the idea behind the implementation of different metaheuristic technologies in anchor-based WSN localization. Similarly, Rayavarapu and Mahapator [15] proposed a Sun Flower Optimization (SFO)-based range-free, anchor-free localization method for WSNs using DV-Hop. It works by forming a new objective function considering energy, distance, and angle to minimize localization error, and demonstrates improved accuracy and convergence compared with conventional heuristic-based localization methods. Hao et al. [16] developed a localization for wireless sensor networks, aimed at overcoming the challenge of accurately determining node positions within these networks, which is considered crucial for applications in healthcare, environmental monitoring, and industrial control. The authors analyse existing localization techniques, classifying them as either distance-based or non-distance-based and outlining their drawbacks. To increase efficiency and accuracy, the research suggests an improved approach that utilises Voronoi diagrams and Support Vector Machines (SVM) for more accurate node positioning. Apart from additive Gaussian noise, non-line-of-sight (NLOS) errors further contribute to localization inaccuracies. Rayavarapu and Mahapatro [17] addressed this problem by utilizing bagging-based ensemble classifiers to identify and resolve NLOS-affected UWB range estimates, as another way of handling outliers besides employing the RANSAC approach used in this work.

Taking standard localization a step further, multiple researchers [18–20] have committed to use trust based parameteres and fuzzy logic to improve localization and routing decisions. Goyat et al. [21] proposed a trust-based localization strategy, combining trust value evaluation based on multiple metrics, with blockchain. This two-step approach provides enhanced protection against various attacks like impersonation attacks,

replay attacks, false location attacks, etc. As complements to trust-scoring methodologies, lightweight authentication and key management techniques were also proposed to secure IoT/fog communications, with the Sarwar et al. [22] scheme to do so, being ECC-based. Hemanand et al. [23] proposed a similar scheme, utilizing K-means clustering for cluster head selection and Mamdani fuzzy inference rule for evaluating the trust score of each node present in the network. The authors made use of the Tversky similarity index to differentiate between the normal and malicious nodes. The performance of the proposed method is analysed over parameters like energy consumption, lifetime of the network, and throughput. Krishnaswamy and Manvin [24] also used a similar trust-based concept, with XoR based encyrption to authenticate nodes present in the network. Fuzzy parameters like node energy, distance, and its mobility are used to calculate the trust score.

The combination of deep learning with Machine Learning (ML) has also emerged as an approach that could provide solutions to security problems. Multiple authors [25–28] have successfully customized IDS with ML/DL-based algorithms for enhanced accuracy and precision. Alserhani [29] proposed integrating metaheuristic algorithms and deep learning to detect malicious nodes in IoT based on blockchain. Gupta et al. [30] demonstrated the potential of deep learning to secure IoT environments by utilizing the Hybrid Chicken Swarm Genetic Algorithm in combination with the MinK-means Algorithm. The authors proved the efficiency of their model on the NSL-KDD dataset. A similar approach using DL was adopted by Hnamte et al. [31], where they evaluated the performance of their proposed model on CICIDS2018 and Edge_IIoT datasets. Wang et al. [32] made use of Whale Optimization Algorithm (PCWOA) to enhance the performance of DV-Hop, with testing done on CEC2013 benchmark. Recently, MOANS DV-Hop was put forward by Rayavarapu and Mahapatro [33], which enhanced the traditional DV-Hop localization in terms of coordinate estimation accuracy with the help of an ideal anchor node subset while removing the range-dependent measurements. Nouman et al. [34] applied machine learning techniques coupled with the blockchain for the detection of a malicious node and distributed data storage. The contribution covers quite valid concerns for identifying security issues, while it remains least focused on integration or application using localization techniques or federated learning. Several authors optimized the results of their proposed models with a variety of feature selection algorithms.

Recent research has also linked CNN-based intrusion detection with dynamic trust management in the context of sensor networks. In their work, Al-Otaibi et al. [35] combined a CNN-IDS with a trust management system that utilized artificial intelligence and a lightweight ECC-based authentication scheme for medical sensor networks, obtaining a good performance score related to detection accuracy in a short time period with minimal energy consumption, even though the trust scores they generated were calculated without concerning localization. Similarly, Khan et al. [36] proposed a trust-based integrity assessment technique for the management of distributed network systems that further confirmed the usefulness of trust assessment incorporated into security monitoring, although without taking account of spatial localization issues. Sarwar et al. [37] conducted a comparative analysis of a number of machine learning architectures, including neural networks applied for anomaly detection in various IoT traffic datasets, which highlights a trend for the use of common intrusion datasets instead of specific attack traces specific to wireless sensor networks a problem that the current work has

in common with CIC-IDS-2017. Talukder et al. [38] enhanced their IDS with the use of SMOTE and XGBoost for vital feature selection. Similarly, authors of [39] enhanced the feature selection of their proposed model with Ant Colony Optimization. Sarwar et al. [40] utilized ML with AI, in identifying network vulnerabilities in the IoT environment of the smart home bedroom while indicating the suitability of ML anomaly detection methods in resource-parallel, ongoing sensor environments like WSN implementations.

Building on previous studies, Gebremariam [41] discussed a concept introduced to protect localization from the degrading quality of services provided by malicious nodes in IoT-based WSN by combining the knowledge of the newly emerging blockchain technology with federated learning. This hybrid scheme uses the desirable characteristics of the distributed ledger technology of blockchain to maintain the integrity of data in training models and to alleviate malicious node activities through federated learning. It supports the goal of improving detection rates with a resulting impact on decreasing localization errors that hold high expectations for the model for future advances in technology in securing the IoT-focused wireless sensor network space.

The review summarised above presents the limitations and research gaps in some of the existing work. Most of the work has failed to address the holistic integration of malicious node detection, energy-efficient clustering, and secure localization within a single unified pipeline. Here, the proposed framework comes into play since it paves the way for a perfect merging of energy-efficient clustering strategies, robust localization schemes, and strong security measures in a manner involving different advanced techniques like DEEC, RANSAC, and GDA, with an IDS powered by Convolutional Neural Network (CNN).

3 Proposed Model

Figure 2 illustrates the system model of the proposed framework, which is divided into four key components:

3.1 IDS at cluster level

IDS plays a very important role in preventing malicious activity against WSNs because such activities might affect the accuracy and reliability of localization data. In pre-localization, an IDS is vital as it will ensure that data fed to the algorithms is not tampered

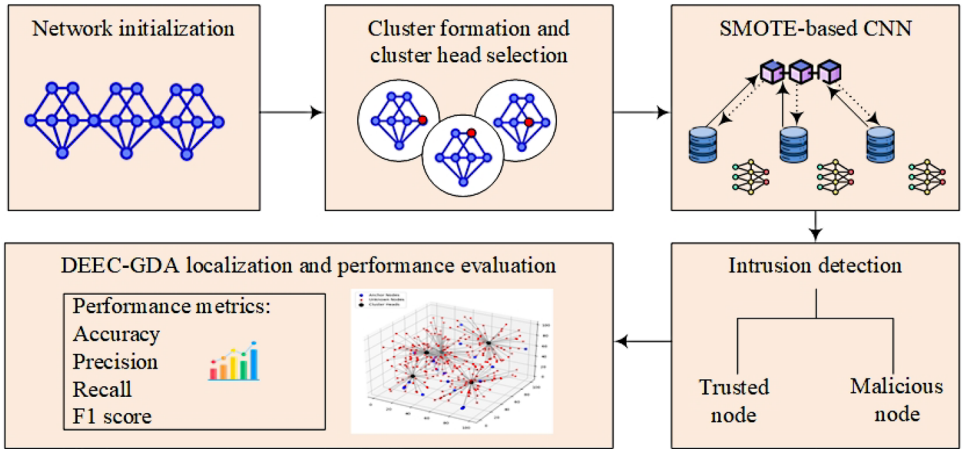


Fig. 2 Structure of the proposed model

with, and against Sybil attacks. Implementing an IDS at this point preserves the integrity of the whole localization process, which means there is less erroneous data being propagated, and the robustness of the network improves. The proposed approach utilises SMOTE-based CNN as the main detection model, offering a lightweight yet powerful deep-learning solution suitable for the constraints and requirements of WSNs. Each CH integrates an IDS layer to detect malicious nodes before proceeding with localization computations. It's important to note that the dataset used is a general-purpose network intrusion dataset and lacks WSN-specific attack signatures. In this research, we have made use of DoS/DDos/PortScan and web attack categories present in the dataset as structural proxies, mirroring the anomalous flow-volume and timing signatures analogous to the traffic irregularities a compromised WSN node would show when injecting falsified data. The use of of a WSN-specific localization-attach dataset is part of the future research of this work. Figure 3 summarizes the steps discussed above.

3.2 Model definition

The model utilises a CNN tailored for binary and multiclass intrusion detection tasks. For the case of binary classification, the architecture begins with an input layer that accepts 9×9 grayscale images, where each image represents a single network flow. The first convolutional layer applies 32 filters of size 3×3 with ReLU activation to extract spatial patterns indicative of benign or malicious traffic. MaxPooling reduces these dimensions, capturing the most significant features and reducing computational complexity.

Each network flow record in CIC-IDS-2017 comprises of 77 numerical features. Followed by Quantile transformer normalization, all the features are ordered by a ranking based on their importance and mapped row wise into a $9 \times 9 = 81$ cell grid. The remaining celss are padded with zero. This transformation preserves the relative importance of the features. The extracted feature maps are flattened into a 1-D vector using a flattened layer, which enables subsequent dense layers to learn high-level representations of the data. A fully connected dense layer of 128 units with ReLU activation processes these features, enabling the model to capture complex relationships in the data. Dropout is used on this layer to prevent over-fitting, as neurons are randomly deactivated during training so that the model generalises well to unseen data. The final output layer classifies the network flow as either benign or malicious. The architecture for multiclass classification is slightly different. The output layer is replaced with a softmax layer, which has neurons equal to the number of classes. This allows for the more accurate assignment of probabilities over multiple categories, allowing for more accurate identification of specific types of attacks. The loss function, sparse categorical cross-entropy, makes the model optimal for multiclass classification tasks. To ensure efficient training, callbacks such as EarlyStopping and ReduceLROnPlateau are employed.

3.3 Fuzzy logic trust inference for node validation

Although the proposed IDS is accurately predicting the intrusion, using it alone may generate false positives/negatives, especially in real-life deployment. Adding a fuzzy trust inference layer refines the decision made by the IDS and ensures that only strictly legitimate nodes are able to participate in the localization. The trust value for each node, i, is calculated by the CH based on three indicators.

The IDS Confidence Score:

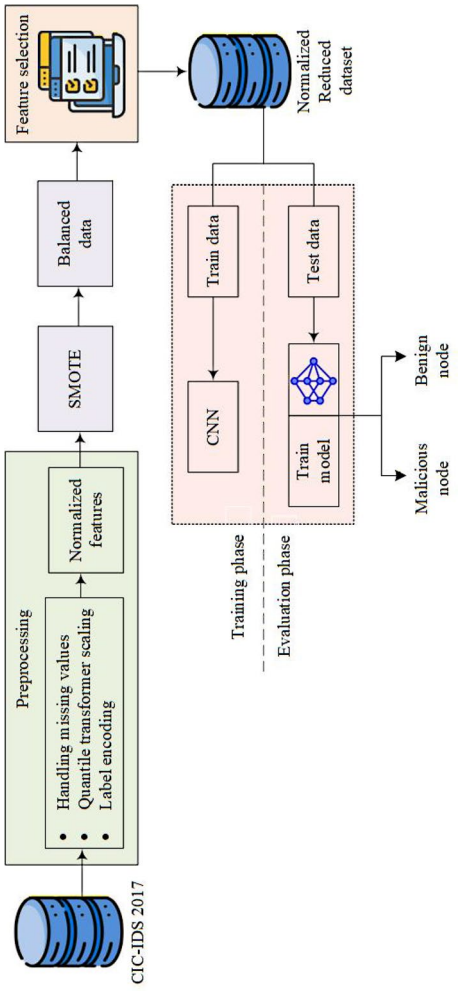


Fig. 3 Preprocessing pipeline details of the proposed IDS model

Algorithm 1 IDS-assisted Fuzzy Trust Secure Localization Framework

Require: Number of nodes N, field size m, initial energy E_0, optimal CH probability P_{opt}, noise standard deviation σ, RANSAC threshold ϵ, trust threshold τ, MAD sensitivity k

Ensure: Estimated positions $\hat{\mathbf{p}}_u$ for all unknown nodes u

Deploy N nodes randomly in $[0, m]^3$ and initialize energies $E_i(0) = E_0$.
Form clusters and select cluster heads (CHs) using DEEC with probability p_i.
for each cluster head CH_k **do**
    Perform IDS inference and obtain malicious probability $\hat{y}_u$ for each node u.
    Compute rough position estimate $\tilde{\mathbf{p}}_u$ via inverse distance weighted centroid of anchor management (Equation 3).
    Compute $S_{ids}(u) = 1 - \hat{y}_u$, $S_{en}(u) = E_u(r)/E_0$, and distance consistency score $S_{dc}(u)$ (Equation 4).
    Apply fuzzy inference and centroid defuzzification to compute trust score $t(u)$.
    Filter nodes: $u \in \mathcal{U}_{valid}$ if $t(u) \geq \tau$.
end for
for each unknown node $u \in \mathcal{U}_{valid}$ **do**
    Collect distance measurements $\tilde{d}_{iu}$ from anchors/CHs.
    Apply RANSAC to remove outliers and form inlier set $\mathcal{I}_u$.
    Apply GDA refinement to estimate $\hat{\mathbf{p}}_u$ using $\mathcal{I}_u$.
end for
Compute localization error metrics and return final estimates $\hat{\mathbf{p}}_u$.

$$S_{ids}(i) = 1 - \tilde{y}_i \tag{1}$$

where $\tilde{y}_i$ is the probability of a node being malicious.

Residual Energy Score:

$$S_{en}(i) = \frac{E_i(r)}{E_o} \tag{2}$$

Distance Consistency Score (RANSAC inlier ratio proxy) Since the RANSAC-based outlier rejection process is applied after trust filtering, the distance consistency score $S_{dc}(u)$ cannot be derived from the RANSAC ratio, without introducing a circula dependency. Thus, to avoid it we $S_{dc}(u)$ is computed via a lightweight consistency check. An estimate on the node position is made via an inverse-distance weighted centroid of its anchor, as illustrated via the Eq. 3.

$$\tilde{\mathrm{p}}_u = \frac{\sum_i w_i \, \mathrm{p}_i}{\sum_i w_i}, \qquad w_i = \frac{1}{\tilde{d}_{iu}} \tag{3}$$

where $\tilde{d}_{iu}$ is the raw measured distance to an anchor node i whose known position is labelled as p_i. The distance predicted using this equation is then compared against each measurement to obtain a residual, $r_i = \tilde{d}_{iu} - \|\mathrm{p}_i - \tilde{\mathrm{p}}_u\|_2$. Using the median (med) and median absolute deviation (MAD) of these calculated residuals, the estimated values are treated as consistent if it falls within k times the MAD of the calculated median residual. The distance consistency score is then calculated as represented in Eq. 4.

$$S_{dc}(u) = \frac{\left|\{i : |r_i - \mathrm{med(r)}| \leq k \cdot \mathrm{MAD(r)}\}\right|}{N_u} \tag{4}$$

where N_u represents the number of anchor measurement available to node u and k represents the sensitivity parameter which can be tuned. Since this requires no

prior outlier-filtered node set, it is computable at the trust-inference stage, before the RANSAC-based refinement is implemented.

3.3.1 Fuzzy membership functions

In the proposed framework, fuzzy membership functions are employed to map the crisp trust indicators into linguistic variables. Each node u is evaluated using three normalized indicators: IDS confidence score $S_{ids}(u)$, residual energy score $S_{en}(u)$, and distance consistency score $S_{dc}(u)$, such that:

$$S_{ids}(u) \in [0,1], \quad S_{en}(u) \in [0,1], \quad S_{dc}(u) \in [0,1]. \tag{5}$$

Each input indicator is represented using three fuzzy linguistic terms:

$$\{\text{Low}, \text{Medium}, \text{High}\}.$$

Similarly, the output trust level $T(u)$ is represented using:

$$\{\text{Low}, \text{Medium}, \text{High}\}.$$

To ensure computational simplicity and interpretability, triangular membership functions are adopted. For a crisp variable $x \in [0,1]$, the triangular membership function is defined as:

$$\mu(x;a,b,c) = \begin{cases} 0, & x \le a, \\ \dfrac{x-a}{b-a}, & a < x \le b, \\ \dfrac{c-x}{c-b}, & b < x < c, \\ 0, & x \ge c, \end{cases} \tag{6}$$

where a, b, and c denote the left endpoint, peak, and right endpoint of the triangular function, respectively.

In this work, the membership functions for each input indicator are parameterized as follows:

$$\mu_{\text{Low}}(x) = \mu(x;0,0,0.5), \tag{7}$$

$$\mu_{\text{Medium}}(x) = \mu(x;0.25,0.5,0.75), \tag{8}$$

$$\mu_{\text{High}}(x) = \mu(x;0.5,1,1). \tag{9}$$

Accordingly, for node u, the membership degrees of IDS confidence are computed as:

$$\mu^L_{ids}(u) = \mu_{\text{Low}}(S_{ids}(u)), \quad \mu^M_{ids}(u) = \mu_{\text{Medium}}(S_{ids}(u)), \quad \mu^H_{ids}(u) = \mu_{\text{High}}(S_{ids}(u)). \tag{10}$$

Similarly, the membership degrees for residual energy and distance consistency are given by:

$$\mu^L_{en}(u) = \mu_{\text{Low}}(S_{en}(u)), \quad \mu^M_{en}(u) = \mu_{\text{Medium}}(S_{en}(u)), \quad \mu^H_{en}(u) = \mu_{\text{High}}(S_{en}(u)), \tag{11}$$

$$\mu^L_{dc}(u) = \mu_{\text{Low}}(S_{dc}(u)), \quad \mu^M_{dc}(u) = \mu_{\text{Medium}}(S_{dc}(u)), \quad \mu^H_{dc}(u) = \mu_{\text{High}}(S_{dc}(u)). \tag{12}$$

These membership functions provide a smooth transition between linguistic categories and allow the fuzzy inference system to handle uncertainty in intrusion detection confidence, energy availability, and measurement consistency during the secure localization process.

3.3.2 Fuzzy rule base

After fuzzification, a rule base is defined to infer each node's trust level using linguistic reasoning. Let the three fuzzy input indicators for a node u be: (i) IDS confidence score $S_{ids}(u)$, (ii) residual energy score $S_{en}(u)$, and (iii) distance consistency score $S_{dc}(u)$. Each input is mapped to the linguistic terms $\{\text{Low}, \text{Medium}, \text{High}\}$, while the output trust level $T(u)$ is described using the same linguistic scale.

The fuzzy rule base is constructed to ensure that a node is regarded as trustworthy only when both security confidence and measurement consistency support its participation in localization. Representative fuzzy rules are defined as follows:

1. **Rule 1:** IF $S_{ids}(u)$ is High AND $S_{dc}(u)$ is High AND $S_{en}(u)$ is High THEN $T(u)$ is High.
2. **Rule 2:** IF $S_{ids}(u)$ is High AND $S_{dc}(u)$ is High AND $S_{en}(u)$ is Medium THEN $T(u)$ is High.
3. **Rule 3:** IF $S_{ids}(u)$ is Medium AND $S_{dc}(u)$ is High THEN $T(u)$ is Medium.
4. **Rule 4:** IF $S_{ids}(u)$ is High AND $S_{dc}(u)$ is Medium AND $S_{en}(u)$ is High THEN $T(u)$ is Medium.
5. **Rule 5:** IF $S_{ids}(u)$ is Medium AND $S_{dc}(u)$ is Medium THEN $T(u)$ is Medium.
6. **Rule 6:** IF $S_{ids}(u)$ is Low OR $S_{dc}(u)$ is Low THEN $T(u)$ is Low.
7. **Rule 7:** IF $S_{ids}(u)$ is High AND $S_{dc}(u)$ is Low THEN $T(u)$ is Low.
8. **Rule 8:** IF $S_{ids}(u)$ is Low AND $S_{dc}(u)$ is Medium THEN $T(u)$ is Low.

3.3.3 Fuzzy inference mechanism

A Mamdani-type fuzzy inference system is employed. The inference process includes: (i) rule evaluation, (ii) aggregation of rule outputs, and (iii) defuzzification to obtain a crisp trust score.

Let $\mu_{ids}^{L}(u)$, $\mu_{ids}^{M}(u)$, and $\mu_{ids}^{H}(u)$ denote the membership degrees of $S_{ids}(u)$ in the fuzzy sets *Low*, *Medium*, and *High*, respectively. Similarly, $\mu_{en}^{L}(u)$, $\mu_{en}^{M}(u)$, and $\mu_{en}^{H}(u)$ denote membership degrees of $S_{en}(u)$, and $\mu_{dc}^{L}(u)$, $\mu_{dc}^{M}(u)$, and $\mu_{dc}^{H}(u)$ denote membership degrees of $S_{dc}(u)$.

For a fuzzy rule R_k, the firing strength α_k is computed using the minimum (min) operator as:

$$\alpha_k = \min\left(\mu_{ids}^{(\cdot)}(u), \mu_{en}^{(\cdot)}(u), \mu_{dc}^{(\cdot)}(u)\right), \tag{13}$$

where $(\cdot)$ indicates the linguistic label employed in rule R_k.

The implication operation truncates the output membership function of the trust set:

$$\mu_{T_k}(z) = \min\left(\alpha_k, \mu_T^{(\cdot)}(z)\right), \tag{14}$$

where $\mu_T^{(\cdot)}(z)$ is the membership function of the trust output fuzzy set associated with rule R_k.

The aggregated output trust membership function is obtained using the maximum (max) operator:

$$\mu_T(z) = \max_k \mu_{T_k}(z). \tag{15}$$

3.3.4 Defuzzification and trust score computation

To obtain a crisp trust score $t(u) \in [0,1]$, the centroid (center of gravity) defuzzification method is applied:

$$t(u) = \frac{\int_0^1 z \cdot \mu_T(z)\, dz}{\int_0^1 \mu_T(z)\, dz}. \tag{16}$$

The trust score $t(u)$ quantifies the reliability of node u, where higher values indicate greater trustworthiness.

3.3.5 Trust-based node filtering policy

After computing the trust score $t(u)$, a threshold-based filtering policy is applied to determine whether node u is eligible to participate in localization:

$$u \in \mathcal{U}_{valid} \iff t(u) \geq \tau, \tag{17}$$

where $\mathcal{U}_{valid}$ denotes the set of valid nodes and $\tau \in [0,1]$ is the trust threshold. Nodes with $t(u) < \tau$ are excluded from subsequent localization computations, thereby mitigating the influence of malicious or inconsistent nodes. The proposed work uses a value of 0.6 for τ.

3.4 Robust localization using RANSAC and GDA

After trust-based filtering, localization is performed using only nodes in $\mathcal{U}_{valid}$. The localization pipeline integrates robust outlier rejection using RANSAC and GDA.

3.4.1 Distance measurement Model

Each unknown node u measures its distance to anchor nodes or cluster heads. Let the true position of node u be $\mathrm{p}_u = (x_u, y_u, z_u)$ and the position of anchor node be $\mathrm{p}_i = (x_i, y_i, z_i)$. The true Euclidean distance is given by:

$$d_{iu} = \|\mathrm{p}_i - \mathrm{p}_u\|_2 . \tag{18}$$

The measured distance is corrupted by additive noise:

$$\tilde{d_{iu}} = d_{iu} + n_{iu}, \tag{19}$$

where:

$$n_{iu} \sim \mathcal{N}(0, \sigma^2). \tag{20}$$

Cluster heads are selected from the anchor nodes, whose positions are known prior to the deployment of this model. In order to avoid circular dependency between the cluster-head position accuracy and the localization process, unknown nodes are never elected as cluster heads.

3.4.2 Outlier detection using RANSAC

RANSAC is applied to remove outlier measurements caused by noise, multipath fading, or falsified ranging information. Let $\tilde{d}_{iu}$ denote the measured distance and $\hat{d}_{iu}$ denote the predicted distance under a candidate model. The residual is computed as:

$$r_{iu} = \tilde{d}_{iu} - \hat{d}_{iu}. \tag{21}$$

A measurement is classified as an inlier if:

$$|r_{iu}| \leq \epsilon, \tag{22}$$

where ϵ is a residual threshold. The inlier set for node u is defined as:

$$\mathcal{I}_u = \{i \mid |r_{iu}| \leq \epsilon\}. \tag{23}$$

Only inlier measurements are forwarded to the refinement stage.

3.4.3 Gaussian distribution approach (GDA) for refinement

After RANSAC filtering, the remaining distance estimation errors are assumed to follow a Gaussian distribution. For each anchor $i \in \mathcal{I}_u$, the likelihood of observing the residual error is:

$$P(e_{iu}) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\tilde{d}_{iu} - \hat{d}_{iu})^2}{2\sigma^2}\right). \tag{24}$$

Assuming independence across anchors, the overall likelihood of the unknown node position p_u is:

$$\mathcal{L}(\mathrm{p}_u) = \prod_{i \in \mathcal{I}_u} P(e_{iu}). \tag{25}$$

Maximizing $\mathcal{L}(\mathrm{p}_u)$ is equivalent to maximizing the log-likelihood:

$$\log \mathcal{L}(\mathrm{p}_u) = -\sum_{i \in \mathcal{I}_u} \frac{(\tilde{d}_{iu} - \hat{d}_{iu})^2}{2\sigma^2} + C, \tag{26}$$

where C is a constant independent of p_u.

Therefore, localization refinement reduces to the nonlinear least squares problem:

$$\hat{\mathrm{p}}_u = \arg\min_{\mathrm{p}_u} \sum_{i \in \mathcal{I}_u} \left(\tilde{d}_{iu} - \|\mathrm{p}_i - \mathrm{p}_u\|_2\right)^2. \tag{27}$$

The solution $\hat{\mathrm{p}}_u$ represents the refined position estimate of unknown node u.

3.4.4 Localization error metric

Localization performance is evaluated using the Euclidean localization error:

$$\mathrm{LE}(u) = \|\mathrm{p}_u - \hat{\mathrm{p}}_u\|_2. \tag{28}$$

The mean localization error over all localized unknown nodes is computed as:

$$\overline{\mathrm{LE}} = \frac{1}{U} \sum_{u=1}^{U} \|\mathrm{p}_u - \hat{\mathrm{p}}_u\|_2, \tag{29}$$

where U denotes the number of localized unknown nodes.

The integration of RANSAC and GDA into the DEEC framework ensures that there would be a smooth operation where there is an interaction and a balance between energy efficiency, robustness in handling noise, and precision of operation. The overall operation is summarized below:

1. Node Deployment: Sensor nodes are randomly distributed in the target area. The nodes that have enough energy are selected as the anchor nodes (CHs) by the DEEC protocol, and the rest remain as unknown nodes.
2. Cluster Formation: The DEEC protocol establishes clusters and selects CHs with residual energy. These CHs act as anchor nodes for localization.
3. Distance Measurement: The unknown nodes estimate their distances to CHs by RSSI or ToA. The measurements have noise and outliers.
4. Outlier Removal using RANSAC: The distance data is processed with RANSAC, where the outliers are discarded, and only reliable measurements are maintained.
5. Localization using GDA: GDA refines the position estimates of unknown nodes by using cleaned distance data from RANSAC. It minimises residual errors and provides confidence intervals.
6. Energy-efficient Communication: DEEC ensures that the refined localization results are communicated efficiently within the network, maintaining the longevity of the system.

4 Results and comparison

This section presents the results achieved after implementing the proposed secure localization framework. In the initial portion of this section, we first describe the experimental setup, followed by the performance analysis of the IDS and the localization algorithm. In the end, a thorough comparison of our approach with existing methods and their clear implications is discussed.

4.1 Experimental setup

Table 1 summarizes the key parameters used in the experimental setup. The experiment was conducted in a simulated 3D wireless sensor network environment. The entire simulation was implemented with Python, using TensorFlow 2.10 for the proposed CNN-based IDS and SciPy for optimization routines.

4.2 Intrusion detection performance

The SMOTE-based CNN IDS was implemented and evaluated using the CIC-IDS-2017 dataset, comprising 2,313,810 samples with 8 real attack categories (DoS, DDoS, PortScan, Web attacks, and more) and 77 features per sample. The dataset was split into training (60%), validation (20%), and testing (20%) sets, with 15 classes (1 benign and 14 representing attack types). The dataset shows a significant imbalance, as nearly 85% of the records belong to the benign category. The use of SMOTE balanced both categories to 1,186,390, where 60% of the dataset is used for training and validation, and testing utilizes 20% each.

In order to avoid data leakage, the quantile transformation was fit on the training split and subsequently applied, without refitting to the validation and test splits. Similarly, SMOTE balancing was applied only to the training data. Model selection using early

Table 1 Experimental configuration

Parameter	Value
Network area	$50m \times 50m \times 50m$
Transmission range	15 m
Total nodes	400
Anchor nodes	60 (15%)
Unknown nodes	340 (85%)
Malicious nodes	34 (10% of unknown nodes)
Initial energy	1.0 J
Measurement noise (σ)	0.5–2.0 m
Trust threshold (τ)	0.6
RANSAC threshold (ε)	2.0 m
Sink Position	$(25\text{m}, 25\text{m}, -10\text{m})$
Transceiver electronics energy(E_{elec})	50nJ/bit
Free-space amplifier energy (ε_{fs})	10 pJ/bit/m^2
Multipath amplifier energy (ε_{mp})	0.0013 pJ/bit/m^4
Data packet size	20,000 bits

Table 2 CNN architecture for intrusion detection

Layer	Type	Output Shape	Parameters
1	Input	$(9, 9, 1)$	0
2	Conv2D + ReLU	$(9, 9, 32)$	320
3	MaxPooling2D	$(4, 4, 32)$	0
4	Conv2D + ReLU	$(4, 4, 64)$	18,496
5	MaxPooling2D	$(2, 2, 64)$	0
6	Flatten	$(256,)$	0
7	Dense + Dropout	$(128,)$	32,896
8	Dense + Sigmoid	$(1,)$	129
Total Parameters			**51,841**

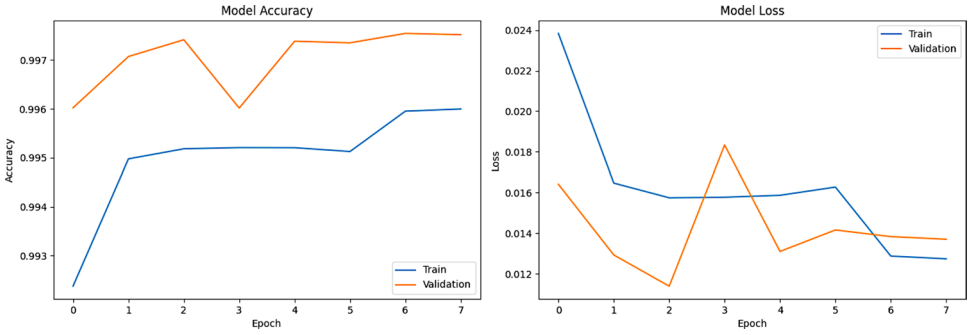


Fig. 4 Training and validation performance of the SMOTE-based CNN model

stopping and ReduceLROnPlateau callbacks monitoring, which prevented the issue of over fitting. Table 2 shows the key parameters of the architecture implemented.

Figure 4 shows the performance of the training and validation, over 10 epochs.

Table 3 presents the comparison of the proposed SMOTE-based CNN IDS with multiple baseline methods, on the same CIC-IDS-2017 dataset. The proposed approach achieves an accuracy of 99.83% with 99.00% precision, 99.84% recall, and 99.42% F1-score. While CNN-1D and DCNN-BiLSTM [31] achieve marginally higher accuracy, these architectures are significantly more complex and computationally heavy. The DCNN-BiLSTM model required 16 epochs of training and more than 6 hours of GPU

Table 3 Performance comparison of intrusion detection methods on CIC-IDS-2017

Method	Accuracy	Precision	Recall	F1-Score
DNN	90.61%	80.85%	84.60%	84.60%
MLP	94.50%	96.80%	96.50%	96.70%
Logistic Regression	87.62%	83.56%	91.60%	84.20%
XGBoost	97.91%	90.30%	96.52%	93.30%
CNN-1D	99.69%	98.79%	99.79%	99.29%
DCNN-BiLSTM [31]	99.81%	99.14%	99.59%	99.36%
Proposed	99.83%	99.00%	99.84%	99.42%

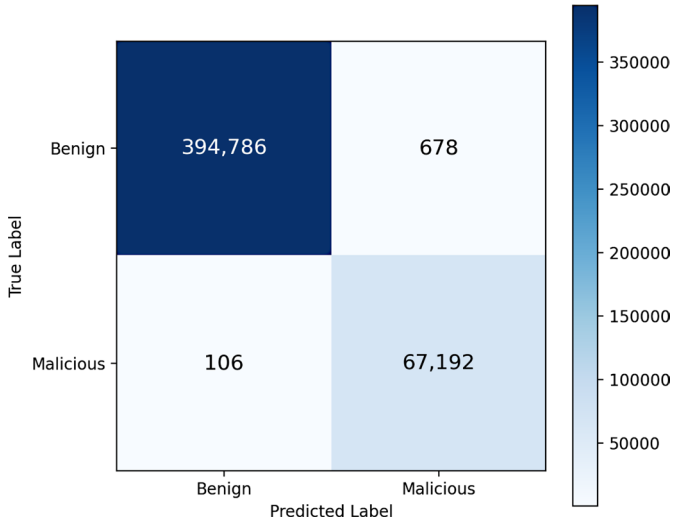


Fig. 5 Confusion matrix showing classification results on the test set

time on the full dataset, whereas the proposed model converged after 8 epochs. More importantly, the proposed model contains only 51,841 trainable parameters, making it adaptable for deployment at resource-constrained WSN cluster heads where memory and latency are vital constraints. The accuracy difference between the proposed model and DCNN-BiLSTM is of only 0.07%, while the proposed model requires substantially fewer computational resources, which is a critical consideration for WSN, where operations are performed on resource-constrained embedded systems.

The confusion matrix shown in Fig. 5 provides a better insight into the classification performance. Out of 462,762 test samples (395,464 benign, 67,298 malicious), the model correctly classified 394,786 benign samples and 67,192 malicious samples and falsely classified only 678 benign samples as malicious. This results in a false positive rate of 0.1714% and a false negative rate of 0.1575%, which is acceptable in a security-driven application where the main focus is to minimize missed attacks.

The low false rate is significant to the proposed TrustLoc Framework, where any missed malicious node at the pre-localization stage could propagate false distances or incorrect coordinate information to the DEEC-RANSAC-GDA stage. The false positive rate of nearly 0 ensures that no legitimate nodes are excluded from the localization stage, preserving the network coverage and the localization density.

4.3 Localization accuracy

After filtering out malicious nodes, we evaluated the localization performance of the DEEC-RANSAC-GDA pipeline. Out of 340 unknown nodes, the proposed framework successfully validated and localized 306 nodes (constituting 90% of the total unknown nodes present in the network). Figure 6 illustrates the localization results in a 3D space. The figure shows the initial network deployment with anchor nodes (blue circles) and unknown nodes (red triangles), along with the final localization estimates (green stars). The close alignment between true positions and estimates illustrates the accuracy of our approach.

The localization error statistics are summarized in Table 4. The mean localization error is 0.458 m with a standard deviation of 0.229 m. The error distribution was slightly right-skewed, with 90% of nodes achieving errors below 0.782 m. The minimum error of 0.056 m demonstrates high precision, while the maximum error of 1.514 m occurs for nodes present at the network boundary or in regions with sparse anchor coverage.

We compared our method against the two existing localization algorithms: DV-Hop and Weighted Hyperbolic (WH). Table 5 presents the results for different numbers of

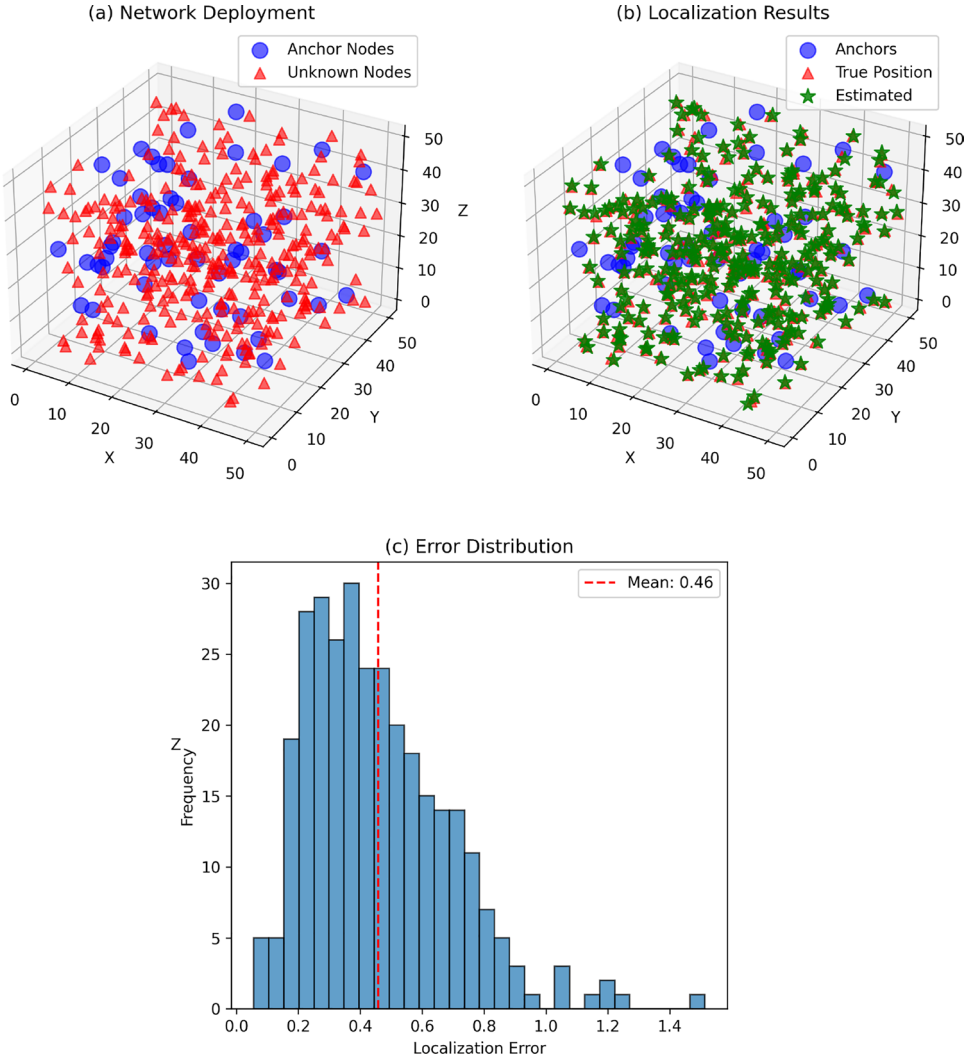


Fig. 6 3D visualization of the WSN showing (**a**) initial deployment of the nodes, (**b**) localization results comparing true positions with estimates, and (**c**) error distribution histogram

Table 4 Localization error statistics (306 validated nodes)

Metric	Value (meters)
Mean error	0.458
Standard deviation	0.229
Median error	0.398
Minimum error	0.056
Maximum error	1.514

Table 5 Comparison of mean localization error (in meters)

Anchor Nodes	DV-Hop	WH	Proposed
20	24.83	0.75	**0.53**
40	22.73	0.51	**0.49**
60	22.22	0.44	**0.46**
80	22.21	0.49	**0.52**

Table 6 Mean $\pm$ and standard deviation of the proposed model over 8 simulation runs

Anchor Nodes	Single-Run Error (m)	Mean $\pm$ std
20	0.53	0.82 ± 0.04
40	0.49	0.56 ± 0.02
60	0.46	0.47 ± 0.02
80	0.52	0.41 ± 0.01

anchor nodes under noise level $\sigma = 1.0m$, showing an approximate 98% error reduction with a mean localization error ranging from 0.46 m to 0.53 m across configurations. The proposed method consistently outperformed DV-Hop and WH across all configurations, except WH at 60 anchor nodes. The stability in the mean error, irrespective of the anchor count, is due to the RANSAC outlier filtering and the Gaussian probabilistic refinement, which effectively handle measurement uncertainties.

4.3.1 Statistical robustness of the anchor-density comparison

The localization error values reported in Table 5 correspond to a single simulation run for each anchor configuration. In order to assess the stability of the reported numbers, the proposed method was also evaluated over eight independent seeded simulation rounds. The mean and standard deviation achieved at each anchor count is further illustrated in Table 6.

At 60 anchors, the gap between the proposed method and Weighted Hyperbolic localization is smaller than the proposed method's standard deviation ($\pm 0.02m$), indicating that this difference is not statistically distinguishable across runs. At 20 anchors, this multi-run experiment yields a higher mean error than a single run value reported in Table 5 which can be attributed to reduced geometric redundancy for trilateration in sparse anchor deployment.

Figure 7 shows the performance across different noise levels and anchor configurations. While WH performs slightly better in very low noise conditions ($\sigma < 0.8m$) with dense anchor deployment, the proposed method demonstrates substantially low error than DV-Hop across all tested noise levels and anchor configurations.

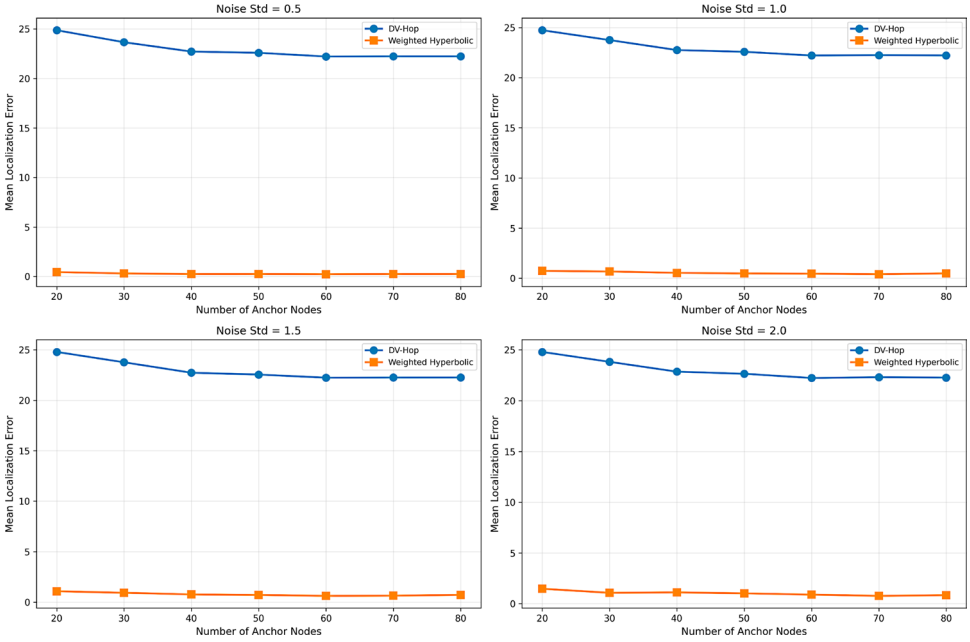


Fig. 7 Localization error comparison across varying noise levels and anchor counts

Table 7 Mean localization error with vs without IDS pre-pipeline stage

Malicious (%)	No IDS (m)	Perfect IDS (m)	Proposed IDS (m)	Improvement
10	2.189	0.461	**0.463**	**78.8%**
20	3.757	0.467	**0.484**	**87.1%**
30	5.487	0.468	**0.477**	**91.3%**

4.4 Impact of IDS pre-filter on localization accuracy

To directly analyse the effect of the pre-IDS stage before the localization, a Monte Carlo simulation experiment was conducted as illustrated in Table 7. The simulation computed 15 independent trials for each malicious node ratio, with each trial randomly deploying 400 nodes across a $50m \times 50m \times 50m$ field. The malicious nodes injected false distance measurements, thereby simulating the environment of a WSN under attack. Three conditions were evaluated across malicious node ratios of 10%, 20%, and 30%. The first condition simulates the effect of no IDS, where all nodes are permitted to participate in the localization process. The second condition simulated the effect of using a perfect IDS, an oracle baseline that removes all the malicious nodes before the localization, and the third condition simulated the proposed IDS, which applies the CNN's obtained false negative rate of 0.1575% and false positive rate 0.1714% from the CIC-IDS-2017 evaluation.

Table 7 and Fig. 8 present the results. Without the IDS, the mean localization error is the highest, reaching 5.487 m at a 30% malicious node ratio. The proposed SMOTE-CNN IDS pipeline reduces this error to 0.477 m at the same malicious ratio, thereby leading to a reduction of 91.3% localization error. Across the entire evaluated attack ratio range, the proposed framework maintains a mean localization error, falling within the range of 0.463 m–0.484 m, thereby proving its stability and reliable attack mitigation across all tested scenarios. Also, the gap between the proposed IDS and the perfect IDS is minimal across all conditions. This performance confirms that the CNN's low false

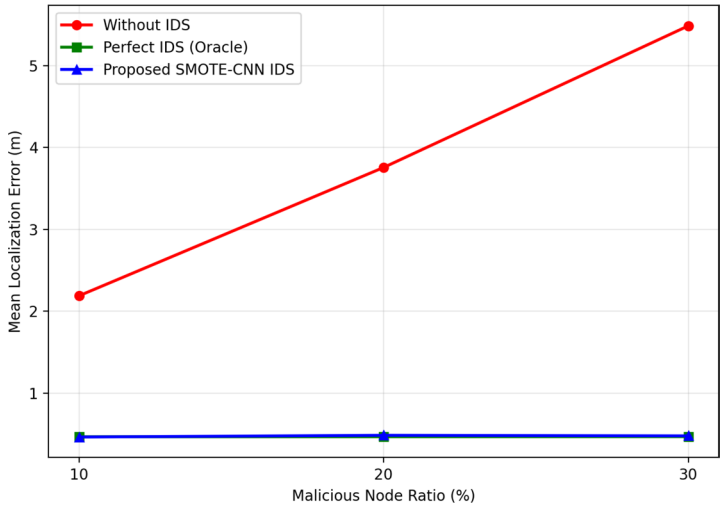


Fig. 8 Impact of IDS in the localization pipeline

Table 8 Ablation study of the proposed localization framework

Configuration	Mean Error	Std
DEEC-only	17.60	0.45
DEEC + RANSAC	18.65	0.39
DEEC + RANSAC + GDA	2.20	0.15
Fuzzy Trust (Full)	0.46	0.01

negative rate has a negligible practical impact on localization quality, even under severe attack conditions.

4.4.1 Ablation study

To analyse the affect of each individual component in the pipeline, we evaluated four independent configurations of DEEC clustering alone, DEEC with RANSAC outlier rejection, DEEC-RANSAC with GDA refinement and the full proposed pipeline. As ellaborated in Table 8 addition of RANSAC alone provides no benefit under the simulated attack model. Adding GDA refinement does provide an improvement, which is further enhanced by fuzzy-trust filtering as proposed in this model. It excludes malicious nodes from reaching the localization stage, thereby reducing the mean error. This confirms that the outlier rejection and the exclusion of the malicious nodes act as complementary components in the proposed localization framework, rather than redundant mechanisms.

4.4.2 Sensitivity analysis of Trust and RANSAC thresholds

The proposed localization framework is dependent on two threshold parameters, the trust threshold τ and ϵ. τ is used in Eq. 17, which decides the node's selection in the localization stage, and the RANSAC residual threshold ϵ, used in Eq. 22 governs the outlier rejection during the position refinement stage. Since both the parameters directly control the trade-off between excluding malicious nodes and retaining enough measurements for accurate trilateration, their impact on localization is evaluated by sweeping $\tau \in \{0.4, 0.5, 0.6, 0.7, 0.8\}$ and $\epsilon \in \{1.0, 1.5, 2.0, 2.5\}$m and the results are elaborated in Table 9.

Table 9 Sensitivity of localization error to the ϵ parameter

	Mean Error (m)
1.0	0.752
1.5	0.572
2.0	0.466
2.5	0.416

Table 10 RANSAC iteration count: accuracy and runtime tradeoff

Iterations	Mean Error(m)	Relative Runtime
30	0.461	1.0×
50	0.443	1.7×
100	0.449	3.3×

The localization error is invariant to τ, across the tested range, due to the computation of trust score based on $S_{ids}(u)$ and $S_{dc}(u)$, which are separated between benign nodes and malicious ones. The errors is considerable more sensitive to ϵ, as represented in the Table 9. Smaller thresholds reject a major ratio of valid but noisy measurements, thus increasing the error by over 60% compared to $\epsilon = 2.0$. The chosen $\epsilon = 2.0$ sits near the diminishing improvement point, from where the improvement ratio moves up by only 11% from 2.0 to 2.5.

The number of RANSAC iterations are kept 30 in this work. This value is selected by comparing the localization accuracy and runtime across 30,50 and 100 iterations, whose results are elaborated in Table 10. As reflected in the table, going beyond 30 iterations failed to increase the accuracy significantly, whereas the runtime scaled.

4.4.3 Robustness to fuzzy membership function

The fuzzy trust inference step maps each input to a {Low,Medium, High} via a triangular membership functions, where breakpoints are fixed at $\{0, 0.5, 1\}$, reflected in Eqs. 7-9. To assess the sensitivity of this parametrization, the baseline breakpoints were compared against a subsection of this range with the rule base and other parameters being fixed. Since this categorization decides how the continuos scores $S_{ids}(u), S_{en}(u)$, and $S_{dc}(u)$ are evaluated by the rule base established in Sect. 3.3.2. All three categorization allowed the identical set of 306 of 340 unknown nodes into the localization stage, producing the same localization error of 0.466 m. This indicates towards the robustness of the node classification the placement of the membership function breakpoints, under the attack model evaluated in this work.

4.4.4 Empirical validation of the ranging error independence assumption

The GDA refinement stage is based on the assumption that the ranging errors $n_{iu} \sim \mathcal{N}(0, \sigma^2)$ are independent of the anchors (Eq. 24). In order to assess the validity of this assumption, the pairwise correlation of ranging residuals across anchors was computed for 300 benign nodes, repeated over 5 independent suimulation seeds. The mean absolute coorrelation came out in the range of 0.045- 0.047, falling close to the expected sampling noise of 0.058 for this sample size. These results are consistent with the independence asumption which forms the base of Eq. 24 and supports its use as a approximation under noisy conditions represented in this work.

Table 11 Node rejection rate and mean localization error versus attack intensity

Malicious (%)	Benign Rejection Rate	Malicious Rejection Rate	Mean Error (m)
0	0.15%	–	0.460
5	0.19%	100.0%	0.467
10	0.04%	100.0%	0.464
15	0.00%	100.0%	0.460
20	0.00%	99.6%	0.489
25	0.25%	99.9%	0.471
30	0.26%	100.0%	0.465

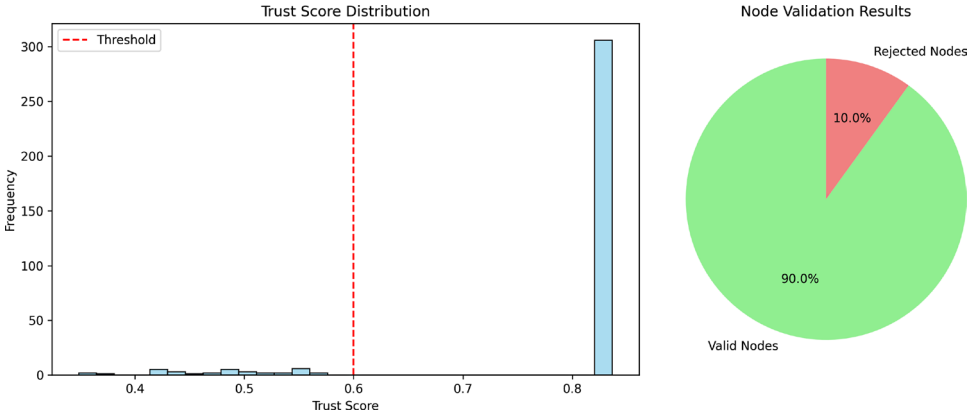


Fig. 9 Trust score distribution with the threshold value of $\tau = 0.6$

Table 12 Energy efficiency comparison

Metric	DEEC	Uniform	Improvement
First node death	452 rounds	321 rounds	+40.8%
Network lifetime	683 rounds	534 rounds	+27.9%
Energy variance	0.121 J	0.343 J	−64.7%

4.5 Security analysis

To assess the framework's resilience under attack, the node rejection rate was checked separately for benign and malicious nodes, where their ratio changed from 0% to 30%.

The proposed IDS-based localization rejects malicious nodes at a nearly 100% rate across the tested range, as illustrated in Table 11. The benign rejection rate stays below 3% throughout the tested range. The mean localization error remains nearly stable, ranging between 0.460 to 0.489, without any significant degradation. These values indicate the framework's resilience to increasing attack intensity, which is majorly due to malicious node rejection rather than threshold discrimination effect specific to moderate attack levels. The trust score distribution in Fig. 9 illustrates the separation between benign and malicious nodes. Benign nodes cluster around trust scores of 0.82–0.86, with a threshold of 0.6, while malicious nodes fall within 0.30–0.60. enabling effective filtering with a false negative rate of only 0.1575% and a false positive rate of 0.1714%.

4.6 Energy efficiency

The DEEC clustering protocol demonstrates significant energy efficiency over uniform random cluster head selection. As shown in Table 12, the proposed approach extended the network lifetime by 27.9%, with the first node death delayed by 40.8%. The adaptive

Table 13 Energy consumption per localization cycle

Component	Energy (mJ/node)	Percentage
RSSI Ranging	12.3	27.3%
IDS Inference	8.7	19.3%
RANSAC	14.2	31.5%
GDA Refinement	6.5	14.4%
Fuzzy Trust	3.4	7.5%
Total	**45.1**	**100%**

Table 14 Runtime breakdown for 400-node network

Component	Runtime (s)
DEEC Clustering	0.19
IDS Inference	0.47
Fuzzy Trust	0.28
RANSAC	4.63
GDA Refinement	1.54
Other	0.84
Total	**7.95**

cluster head selection based on residual energy ensures balanced energy consumption across the network, reducing the energy variance by 64.7%. The energy consumption breakdown for each localization cycle is presented in Table 13. RANSAC outlier detection consumes the most energy (31.5%), followed by RSSI ranging (27.3%) and IDS inference (19.3%).

4.7 Scalability performance

The computational efficiency of the framework was evaluated by measuring the runtime of each component present in the proposed model. The experiments were conducted on a 4-core Intel Xeon CPU with 31GB RAM. Table 14 shows that the complete localization pipeline consisting of 400 nodes requires 7.95 seconds to execute. RANSAC requires the maximum runtime of 58.3%, followed by GDA refinement taking 19.4% of the total runtime. The IDS inference, takes only 0.47 seconds. The framework proves to be highly scalable by achieving nearly a linear scaling in runtime when tested over nodes ranging from 200 to 1000 nodes. Over a network of 1000 nodes, the total localization time comes out to 18.4 seconds. The per-node localization time remains constant at approximately 18 ms, confirming the scalability of the proposed approach.

4.8 Complexity analysis

The per-node computational cost analysis of each inidividual stage in the pipeline is elaborated below, where N denoted the number of unknown nodes, A is the number of anchor nodes available and R represents the number of RANSAC iterations.

- CNN-IDS inference: Given a fixed architecture 2,the inference cost per node is constant, thereby yielding $O(N)$.
- Fuzzy trust inference: Membership evaluation, rule firing and centroid defuzzification requires a fixed number of rules and triangular membership functions per node, thereby yielding $O(N)$ time complexity.

- RANSAC outlier rejection: Each RANSAC iteration, uses a minimal set of anchors and evaluate residuals against all A, giving $O(A)$ per iteration and $O(R \cdot A)$ per node. Expanding this to all trust-validated nodes, this stage will require $O(N \cdot R \cdot A)$.
- GDA refinement: The nonlinear least-squares refinement uses an iterative Gauss-Newton solver which is bound by a fixed iteration count, giving $O(A)$ per node and $O(N \cdot A)$ in all.
- DEEC clustering: Cluster head selection is $O(N)$ and assignment of unknown nodes to the nearest CH is $O(N \cdot C)$, where C represents the number of active CHs.

Combining the components specified above, overall the per-round complexity is $O(N \cdot R \cdot A)$, given the dominant RANSAC stage.

5 Conclusion and future work

This work proposes TrustLoc, a hybrid secure localization framework for wireless sensor networks by integrating energy-efficient clustering, RANSAC-based outlier suppression, Gaussian error modeling, a Mamdani fuzzy trust inference system, and a lightweight SMOTE-based CNN intrusion detection at the pre-localization stage. Evaluated on the CIC-IDS-2017 benchmark dataset comprising 2,313,810 network flow records, the proposed IDS achieves an accuracy of 99.83%. End-to-end simulation demonstrates the efficiency of the pre-localization IDS by reducing the mean localization error by up to 88.2% when compared to an unprotected environment with 30% malicious nodes ratio. The DEEC clustering protocol extends the network lifetime by 27.9% under uniform cluster head selection. Localization error is reduced to a mean of 0.46 m–0.53 m compared to DV-Hop's 22–25 m across varying anchor configurations, confirming the robustness of the RANSAC-GDA pipeline. Although the presented framework shows promising outcomes, various improvements can be obtained by incorporating other security mechanisms, including homomorphic encryption, to enhance data privacy in IDS and localization systems. The use of deep reinforcement learning could make the system more adaptive to the current network conditions.

Author contributions
A.S and R.S wrote the original draft and contributed to the methodology and result analysis. S.D. and S.Sharma and S. Solanki contributed to the figures and literature survey with assistance from P.R., who revised the final document and validated the results.

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Data availability
The dataset analysed during the current study is available at https://www.unb.ca/cic/datasets/ids-2017.html repository.

Declarations

Ethical approval
Not applicable.

Consent to participate
Not applicable.

Consent for publication
Not applicable.

Generative AI and AI-assisted technologies in the writing process
The author made use of the Grammarly tool to increase the quality of the text and to ensure the use of correct grammar throughout the research paper.

Competing interests
The authors declare no competing interests.

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